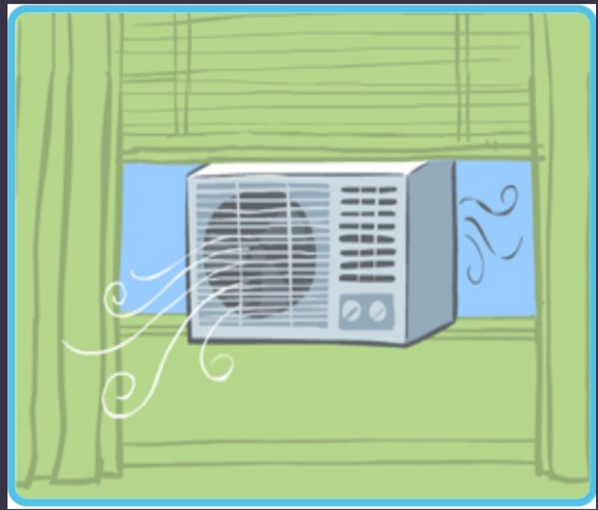
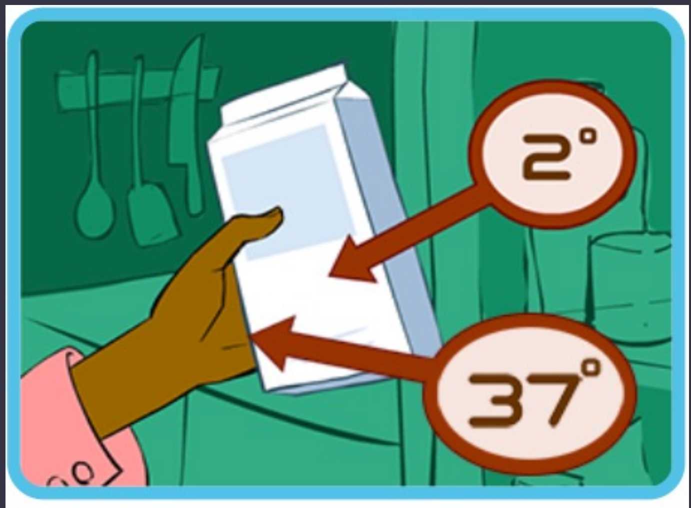
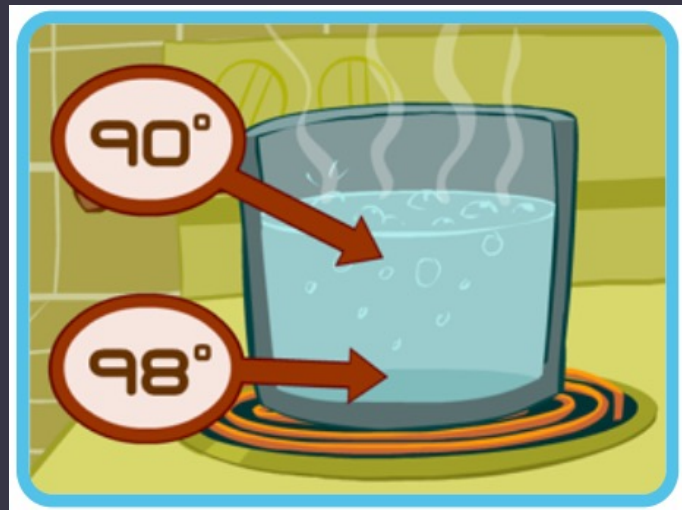



**Thermal energy** is the energy in a system that determines the system's temperature.

Everything has thermal energy.

We feel or see thermal energy only when it **transfers**, or moves, from one place to another.



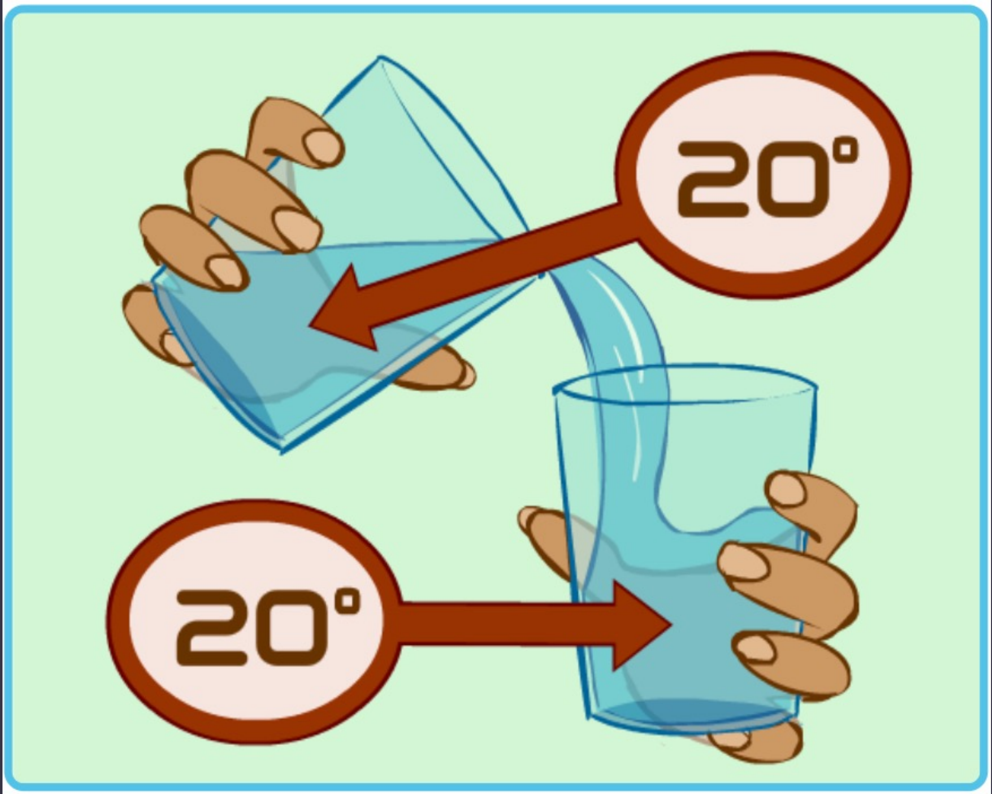

**Thermal energy transfer** occurs only when there is a difference in temperature.

Thermal energy moves from a place that has a higher temperature to a place that has lower temperature. It moves from a hotter place to a cooler place.

Is there thermal energy transfer?

NO

Why?




This picture does not show energy transfer because the temperature of the water is the same in each glass.

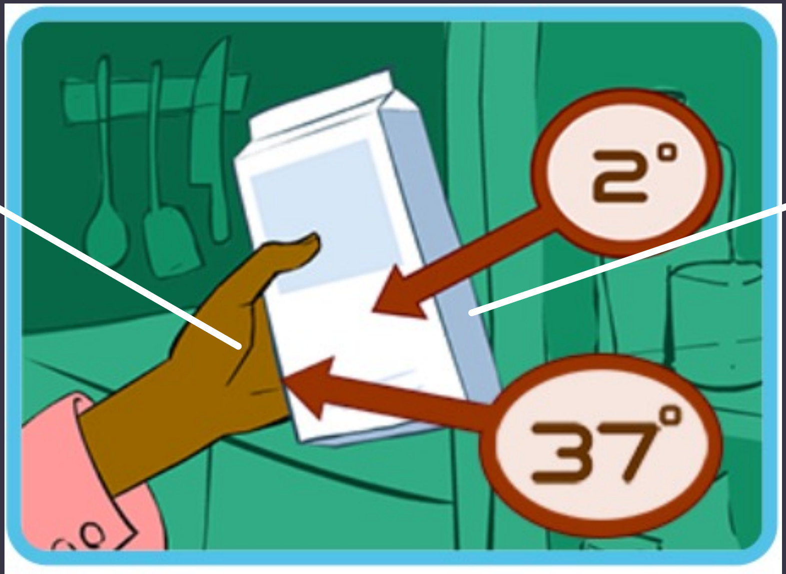
The place with the higher temperature is called the **source**.

The place with the lower temperature is called the **destination**.

Hand

higher temperature

Source




Milk Carton

Lower temperature

Destination

An object's **state of matter** affects how thermal energy is transferred between it and other objects.

When a source and destination are touching, and at least one is a solid, the thermal energy transfer is called conduction.

Source:

stove

State of Matter:

solid

Destination:

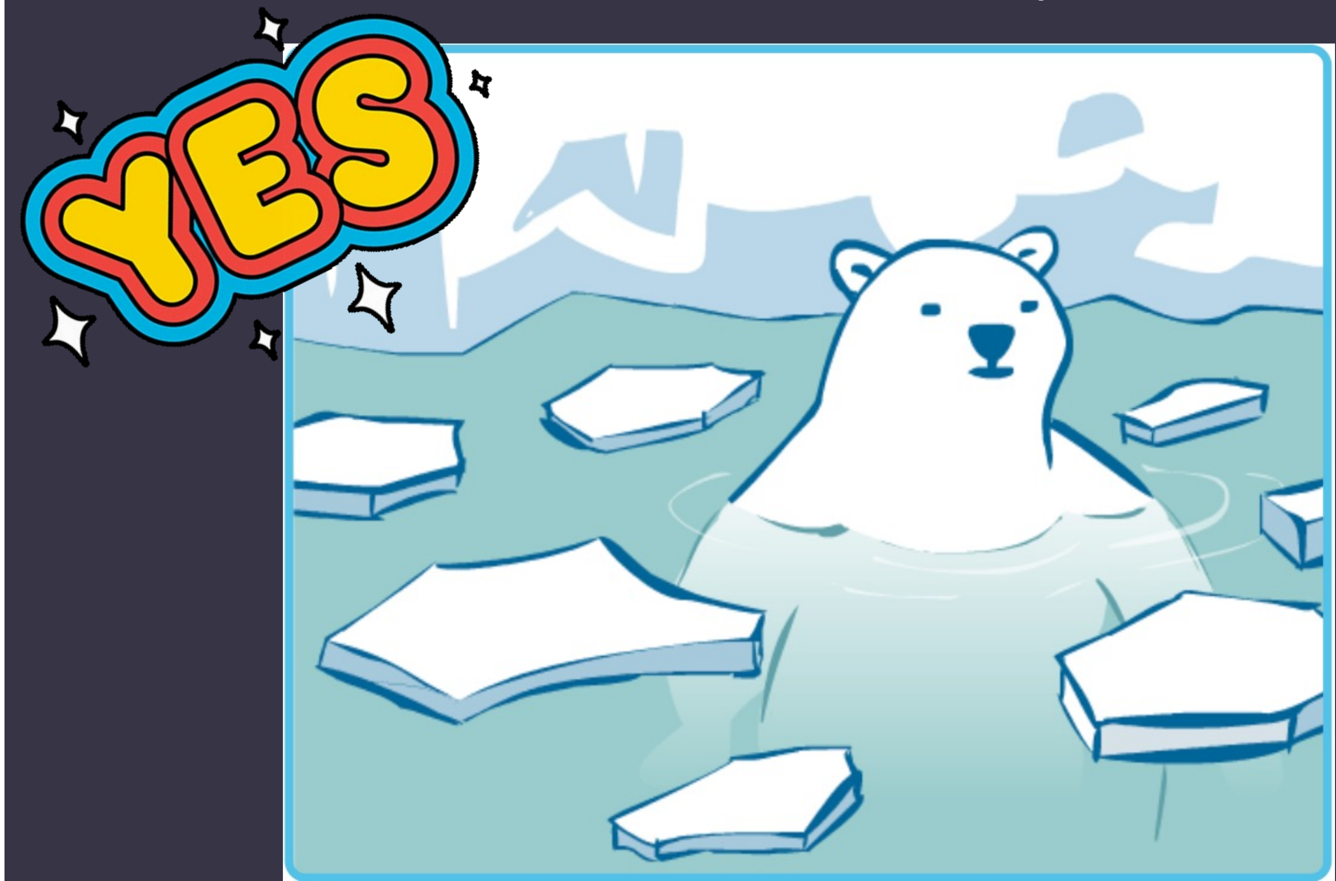
Bottom of the pot

State of Matter:

solid

The water in this picture is warmer than the ice, so the ice is melting.

Is this an example of conduction?




The source is the...

water

The destination is the...

ice

The water and ice are...

touching

Heat is moving through this metal rod.

What is the source of thermal energy in this picture?


A.) The tip of the metal rod

B.) The handle of the metal rod.

How do you know?

A.) The tip of the metal rod is solid.

B.) The tip of the metal rod is lower.

C.) The tip of the metal rod is hotter.

D.) The tip of the metal rod is cooler.

This spoon is in a pot of hot water, which is getting hotter.

Is this an example of conduction?

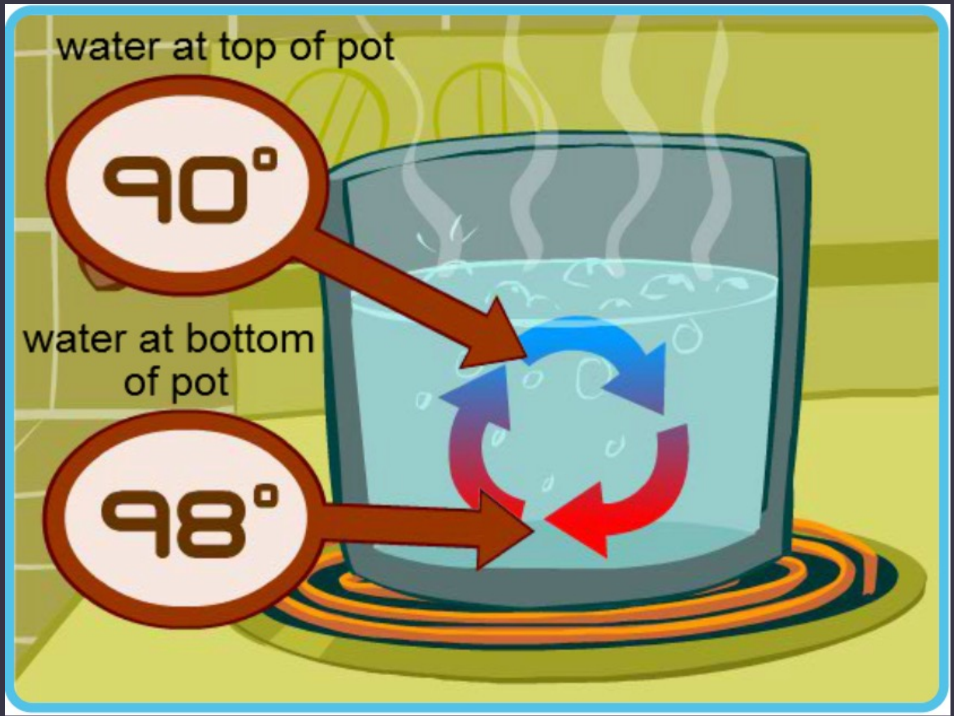

Which statement about conduction is true?

A.) Both objects must be solids.

B.) At least one object must be solid.

C.) Neither object can be a solid.

Thermal energy is being transferred within this pot. The water at the bottom of the pot heats up and moves to the top of the pot.




The water at the bottom of the pot is...

... the source.

The water at the top of the pot is...

... the destination.

State of matter of the source:

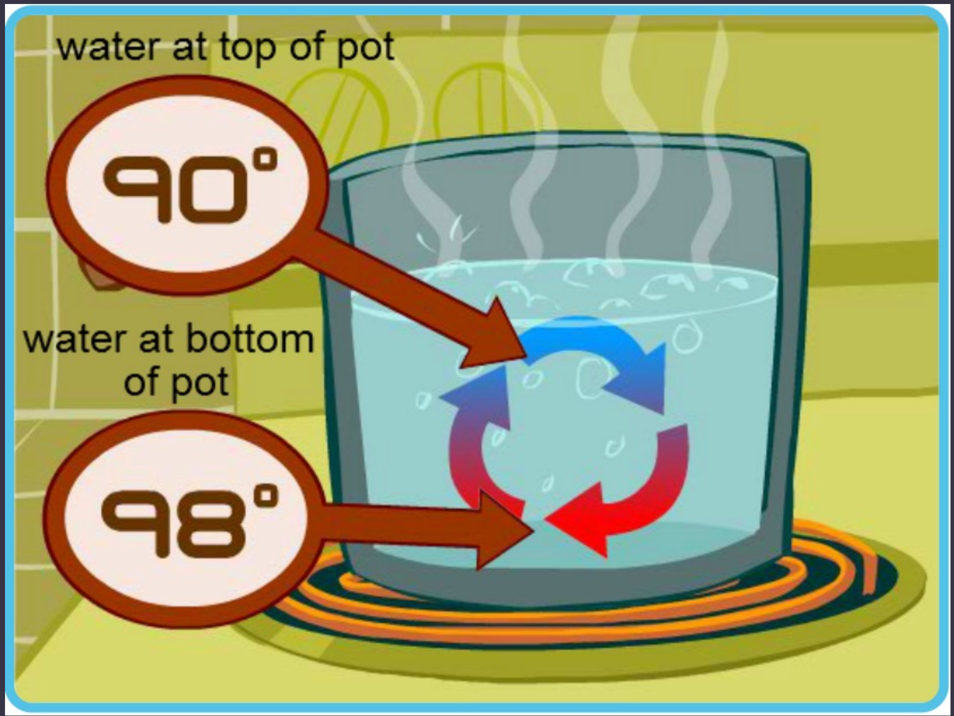
Liquid

State of matter of the destination:

Liquid

**Convection** is the transfer of thermal energy by movement within a liquid or gas (called a fluid).

The warmer parts move up and the cooler parts move down.




How do you know this image shows an example of convection?

Thermal energy is being transferred within a liquid.

- An object’s **state of matter** affects how thermal energy is transferred between it and other objects.
- When a source and destination are touching, and at least one is a solid, the thermal energy transfer is called conduction.
- **Convection** is the transfer of thermal energy by movement within a liquid or gas (called a fluid).

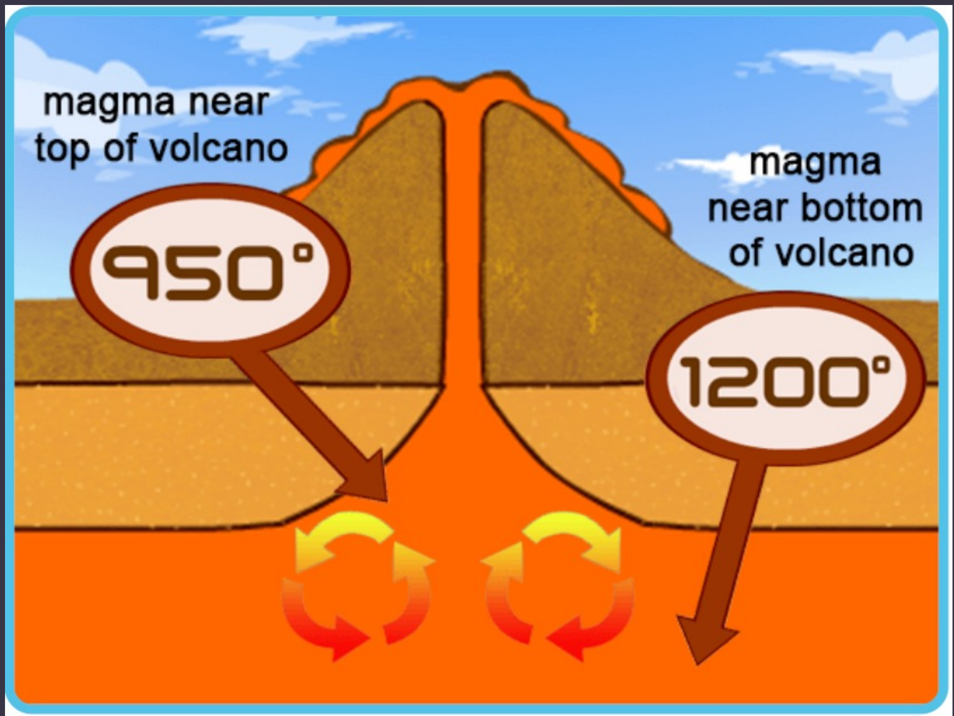
  The warmer parts move up and the cooler parts move down.

| Thermal energy transferred... | | | | |
|---|---|---|---|---|
| solid to liquid | liquid to liquid | gas to gas | solid to solid | solid to gas |
| Conduction | Convection | Convection | Conduction | Conduction |

Thermal energy transfer occurs inside a volcano.

The magma near the bottom of the volcano moves... up.

The magma near the top moves ... down.




The magma near the bottom of the volcano is the... source.

The magma near the top of the volcano is the... destination.

The magma in the volcano is a liquid. This type of thermal energy transfer is called...

convection.

- In conduction and convection, thermal energy moves through **particles** that make up solids, liquids and gasses.
- Not all thermal energy transfer requires particles of matter.
- Radiation moves in waves that can travel through a vacuum, a place with no matter.
- **Particles** of matter heat up when they absorb radiation.

What is the source of radiation in this image?

The **sun** is the source of radiation.

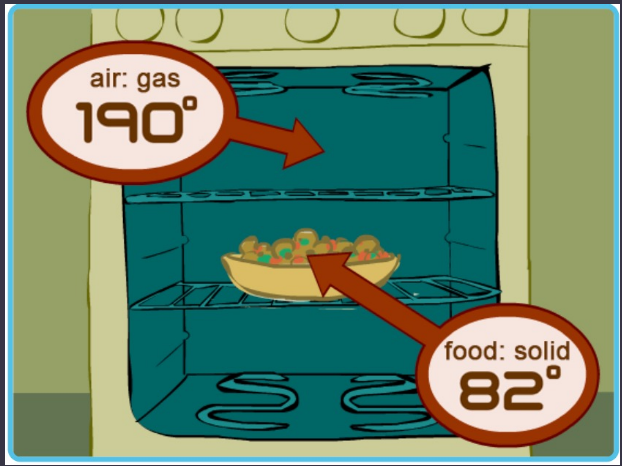
The light you see from a fire is radiation.
Some of the heat you feel from a fire is also radiation.
The fire is a source of thermal energy transfer.

What is the destination of the radiation in this image?

The hands are the destination of the radiation.

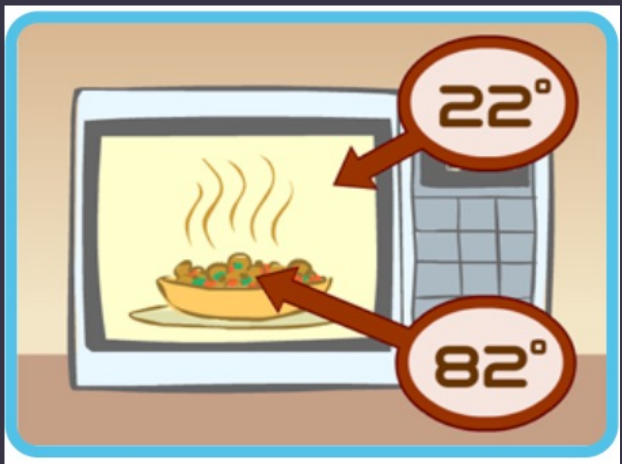
Do these objects need to be touching for radiation to occur?

NO



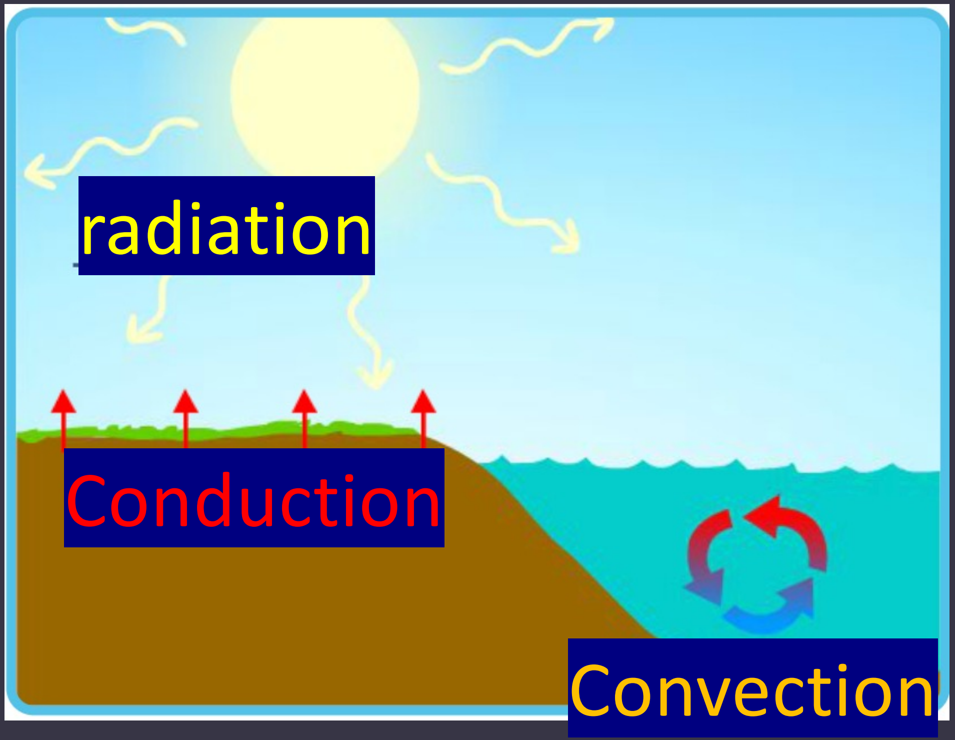

A regular oven heats the air around the food. The hot air cooks the food through conduction.

Microwave ovens make waves that cook the food directly.




Inside the regular oven, the air is _hotter_ then the food.

Inside the microwave oven, the air is _cooler_ than the food.




Conduction, convection and radiation all happen on earth.
In this image, energy moves from the sun to the earth’s surface, warming the soil and the water. The warm soil heats the air above it. The warm water mixes with the cool water in the ocean.

Name the forms of thermal energy transfer in this image.